Multimodal Satellite-Field Data Integration for Scalable Crop-Type Mapping With Limited Labels

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***Abstract*— Crop-type mapping is crucial for improving crop yield predictions, informing agricultural policy decisions, and addressing the impacts of climate change. Despite achieving state-of-the-art performance, current transformer-based architectures that use Satellite image time series (SITS) require costly labeled data and struggle to generalize to new regions. To address this limitation, this article presents a method that leverages multimodal information from field images and satellite data to address the scarcity of reference data in crop-type mapping. The proposed approach consists of three main steps: 1) multimodal data preparation; 2) multimodal pseudo-labeling; and 3) SITS-based crop-type mapping. The first step processes both data sources for facilitating their integration, while the novelty lies in the second step, where additional pseudo-labels are generated by classifying crop parcels with both SITS and field image availability, leveraging multimodal information to produce reliable label predictions. Finally, the third step produces a crop-type map for the entire study area using the SITS and the enriched training set. Experimental results at European level, using the 2018 land use and land cover survey (LUCAS) field images paired with Sentinel-2 SITS, demonstrate the effectiveness of integrating satellite top-view data with ground-level field images. The pseudo-labeled data achieves performance comparable with models trained on the complete labeled dataset, while substantially reducing the need for extensive labeled data collection.**

***Index Terms*— Deep learning (DL), field images, fusion, ground level, land use and land cover survey (LUCAS), multimodal data, satellite image time series (SITS), Sentinel-2.**

I. Introduction

LARGE-SCALE crop-type maps are a key tool for managing agri-food systems. They are used to estimate essential agricultural variables (EAVs) and have already been deployed for applications like the allocation of farm subsidies. In the face of the upcoming challenges on agri-food systems due to environmental or demographic changes, efforts have been made toward mapping cropland on a large scale using Earth observation (EO) data [1], [2], [3]. The advent of the recent Sentinel-2 satellite mission of the European Copernicus Programme significantly contributed to agricultural mapping and monitoring. Thanks to their high-temporal frequency and spatial resolution, Sentinel-2 data have been used to generate irrigation and cereal maps at the global level [4]. Crop cultivation patterns vary significantly from year-to-year, necessitating frequent updates to monitor agricultural practices [5]. To address this challenge, initiatives like the ESA WorldCereal project have focused on developing seasonally updated crop-type maps [4].

Crop-type mapping is framed as *parcel-based classification* when boundaries are known [6], an easier task as input pixels are already in homogeneous clusters, or *semantic segmentation* when boundaries are not available, prediction is pixel-level using contextual spatial information. *Panoptic segmentation* [7], which aims to also retrieve parcel boundaries while performing segmentation, is the most challenging task formulation. To support these more challenging tasks, specific benchmarks have been proposed, such as [8] and [9]. Recently, many specialized deep learning (DL) architectures have been introduced for Satellite image time series (SITS) [10]. For *parcel-based classification*, the pixel-set encoder with temporal attention encoder (PSE-TAE) [11] is a landmark contribution that adapted principles of the canonical transformer [12], by directly using the input tokens as values and defining a single main query for the given sequence. It earned widespread adoption [5], [13], [14], [15], [16], preceding implementation improvements, with a lightweight [17] and a 2-D version [18], and an adaptation to segmentation [7]. Also, for classification and segmentation, Tarasiou et al. [19] proposed the better performing but heavier, temporo-spatial vision transformer (TSViT), a fully attention-based architecture that reversed the commonly adopted spatial-then-temporal factorization. Finally, recent efforts are shifting toward self-supervised learning (SSL). Notably, using an objective inspired by BERT, Dumeur et al. [20] proposed a U-Net to process the spatial dimension and a transformer block for the temporal dimension.

Although these architectures can be highly effective in generating crop-type maps using SITS, they require a large amount of labeled data for training and validation. The necessity of having sufficient labeled data that accurately represents the entire study area is even more evident when working at the country or continental scale, as these models still face challenges in generalizing across new spatial contexts, e.g., different continents or biomes [21]. Finally, even though EO-based DL models have proven effective in generating accurate crop-type maps, the performance of EO-based DL often deteriorates when applied to different

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temporal contexts (e.g., across years). Specifically, due to factors like crop rotation practices and variations in crop phenology, a DL model that performs well for a particular year may exhibit a significant drop in accuracy when applied to another year [22]. This limitation underscores the need for new reference data to fine-tune models effectively for each specific temporal scenario. To mitigate the lack of reference data, efforts have focused on label-efficient methods [10]. Transfer learning and "few-shot" methods have been widely investigated in the EO literature. The process of fine-tuning a supervised model trained in the source domain (i.e., a study area where labeled data are available), usually using fewer labeled samples in the target domain (i.e., a different study area spatially disjoint from the source domain) is referred to as transfer learning and has successfully been applied for crop mapping applications [23], [24]. Few-shot learning involves a similar process but uses dramatically fewer labels in the target, which is helpful to handle less frequently occurring crop types [25]. Domain adaptation involves additional methods to mitigate the distribution shifts between domains, which otherwise affect performance on the target [26]. Finally, much attention has been paid to unsupervised domain adaptation (UDA) methods to address scenarios where no target labels are available [27], [28].

Recent works demonstrated the potential of photo-interpretation of field images, i.e., collected in situ, to extract relevant crop information [29]. Using field images in agricultural applications is not new, and DL methods have been proposed to address fine-grained tasks from disease detection to postharvest quality control [30]. However, for crop-type mapping, they have mainly been used to generate labels for the supervision of models that operate on satellite data [31], [32], [33], [34], [35]. There is a rich body of literature on integrating ground-level images and satellite data [36]. In particular, many approaches have explored cross-view geolocation, e.g., [37], and use geometric properties and reconstruction objectives to align the two perspectives before identifying correspondences for location. However, they typically require high-resolution remote sensing (RS) data such as aerial images. Choumos et al. [38] introduced decision-level fusion from Sentinel-1 and Sentinel-2 and street-view images for binary grassland classification in The Netherlands. Recently, Liu et al. [39] used street-view images to generate geotagged crop-type labels with associated uncertainty and PlanetScope satellite images to derive the corresponding field boundaries. Recent works also pair ground images with single satellite acquisitions for self-supervised pretraining [40]. This pretraining method was tested on a crop-type mapping downstream task using aerial images instead of satellite images in [41].

Unlike previous works that generate pseudo-labels from ground images alone, this article proposes multimodal pseudo-labeling, which aims to exploit the multimodal information provided by field images and satellite data to address the scarcity of reference data in crop-type mapping. The method has three main steps: 1) multimodal data preparation; 2) multimodal pseudo-labeling; and 3) SITS-based crop-type mapping. The first step prepares both data sources for seamless integration. In contrast, the second step generates an additional set of pseudo-labels to mitigate the scarcity of reference data by utilizing the complementary information from field images. Finally, the third step produces a crop-type map for the entire study area by combining the information from the SITS and the pseudo-labeled training set.

A. Motivation

Geotagged field images offer a promising alternative to sharply speed up in situ data collection based on manual annotation. Due to the rise in crowdsourced data, there has been a growing interest in understanding how to use field images to create reference data for different applications [42]. However, an operational solution still lacks to convert these images into usable information. Current approaches primarily focus on confirming farmer-based declarations and are tested on small datasets representative of specific study areas [43]. While combining SITS with field images shows potential, this integration is still exploratory in agricultural mapping and has only been tested in limited spatial contexts. In this framework, an urgent need exists to explore the potential interactions between the two modalities for crop mapping and quantitatively assess integrated modeling approaches. Several operational and technical challenges remain unresolved when working with multimodal data. Specifically, there is a lack of clear guidelines on which types of field images are most effective for integration with satellite imagery, as well as on the DL models that provide the most accurate results when applied to agricultural field images. In addition, despite the growing interest in multimodal approaches within the EO community, no scalable solution has been proposed for integrating field images with SITS that can effectively handle complex and diverse agricultural landscapes.

B. Contribution

The first contribution of this work is the evaluation of different field image collection protocols. For agricultural studies, two types of images are typically collected: "close-up" images [44], [45], which capture detailed crop features such as leaves, bark, flowers, or fruits, and "landscape" images [42], [46], which provide a broader spatial context. Both modalities are compared to determine the most effective agricultural field image collection approach.

The second contribution of this article is to provide a systematic benchmarking of DL models of varying sizes and configurations to classify field images. This evaluation focuses on understanding the robustness and scalability of these models for downstream agricultural mapping tasks, emphasizing their applicability in diverse agricultural settings. Moreover, state-of-the-art multimodal fusion strategies will be tested and compared with develop a robust and scalable solution for agricultural mapping.

Finally, this work seeks to establish a comprehensive multimodal framework to streamline reference data generation, reducing the need for traditional, labor-intensive field surveys. Supervised classification of remote sensing data depends on the availability of large, high-quality annotated datasets. However, generating such datasets through photo interpretation is resource-intensive and time-consuming, even when supported

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by high-resolution satellite imagery. These limitations become particularly significant when scaling to national or continental levels, where efficient use of extensive remote sensing archives is crucial. Collecting field data are even more expensive and poses greater logistical challenges. The main objective of our approach is to demonstrate that leveraging the multimodality of field images and satellite data can expand existing training datasets for a given target year. Given the growing availability of satellite data and both labeled field images and unlabelled street-level images, the proposed method has the potential to enable continuous production of dynamic, seasonally updated reference data, supporting more adaptive and accurate crop mapping systems.

II. Problem Formulation and Notation

In this section, we define this article 's notation and formalize the multimodal crop-type mapping task. In general, the dataset $\mathcal{D} = \{(\{X_i^{(m)}\}_{m\in\mathcal{M}}, y_i)\}_{i=1}^N$ represents is associated with a set of N geotagged parcels each associated with inputs $X_i^{(m)}$ from the set $\mathcal{M}$ of available modalities and the corresponding crop label $y_i \in \Omega = \{\omega_k\}_{k=1}^K$. In particular, we consider a multiview setting where each parcel is associated with an SITS $X^{(S)}$ and the corresponding field image $X^{(P)}$. The SITS is represented as $X^{(S)} \in \mathbb{R}^{T\times C\times H\times W}$ with T is the temporal, C is the spectral, and H and W are the spatial dimensions, respectively. A natural image is represented as $X^{(P)} \in \mathbb{R}^{C'\times H'\times W'}$. It is worth noting that the field image $X^{(P)}$ is either acquired according to the "close-up cover" or "landscape" protocol. For simplicity, we consider multiclass object-level classification as the downstream task, thus predicting the crop label among the closed set of K classes per parcel. This task can be decomposed into a two-step process. First, a unimodal encoder $f^{(m)} : X^{(m)} \to z^{(m)}$ extracts high-level feature representations for each modality $m \in \mathcal{M}$. Similarly, a multimodal encoder $g : (X^{(S)}, X^{(P)}) \to z$ produces a joint representation. Then, a prediction head h takes a unimodal representation $z^{(m)}$ or a multimodal representation z and maps it to the final prediction $h : z \to \hat{y}, \quad \hat{y} \in \mathbb{R}^K$. Finally, F is the entire end-to-end network that maps a set of input modalities $\{X^{(m)}\}_{m\in\mathcal{M}}$ (which can be unimodal or multimodal) to the final prediction: $F : \{X^{(m)}\}_{m\in\mathcal{M}} \to \hat{y}, \quad \hat{y} \in \mathbb{R}^K$.

III. Multimodal Crop-Type Identification

Fig. 1 shows the proposed multimodal pseudo-labeling approach for crop-type mapping with limited labeled data. The method is made up of three main steps: 1) multimodal data preparation; 2) multimodal pseudo-labeling; and 3) SITS-based crop time mapping. In the following, details are given.

A. Multimodal Data Preparation

The first step of the method aims to prepare both data sources for integration, optimizing their information content to achieve the best possible crop-type classification results and a prerequisite for training the initial models that will later derive pseudo-labels. Since the downstream task involves patch-level classification, field images are paired with SITS based on their location and acquisition time. Specifically, each field image is paired with an SITS patch centered at the same coordinates, such that the acquisition timestamp of the field image falls within the temporal range of the SITS acquisitions during the considered year. After preprocessing, these spatiotemporally aligned pairs directly feed into training the multimodal classifier Section III-B that generates pseudo-labels. For each sample, the field image provides a detailed representation of the crop's status, while the SITS captures the evolution of crop phenology. Each crop parcel i is associated with a raw SITS patch $X_{i,\text{raw}}^{(S)}$. The cloud cover is first masked according to the metadata provided with the considered satellite data. Then, each spectral band of the SITS is clipped using the 0.05 and 0.95 quantiles (1). The data are normalized by rescaling the spectral values according to formula (2), where $\mathbf{m} \in \mathbb{R}^C$, $\mathbf{q}_{0.95} \in \mathbb{R}^C$, and $\mathbf{q}_{0.05} \in \mathbb{R}^C$ represent the median, 95th percentile, and 5th percentile values, respectively, computed per spectral band across the dataset. These precomputed vectors are broadcasted across the spatial and temporal dimensions of $X_{i,\text{clip}}^{(S)}$ for normalization. Finally, an SITS of 12 monthly composites is generated by calculating the median value of the acquisitions collected each month for each spectral band [47], i.e., $X_i^{(S)} \in \mathbb{R}^{12\times C\times H\times W}$. A standard interpolation method is applied to fill in missing spectral information caused by heavy cloud coverage by replicating the spectral value from the nearest monthly composite. While still capturing the dynamic changes in crop phenology throughout the growing season, the SITS of monthly composites provides a harmonized dataset across the whole study area from the temporal viewpoint

$$X_{i,\text{clip}}^{(S)} = \text{clip}_{\mathbf{q}_{0.95},\mathbf{q}_{0.05}}(X_{i,\text{raw}}^{(S)}) \tag{1}$$

$$X_{i,\text{norm}}^{(S)} = \frac{X_{i,\text{clip}}^{(S)} - \mathbf{m}}{\mathbf{q}_{0.95} - \mathbf{q}_{0.05}}. \tag{2}$$

To minimize the impact of data quality on model performance, minimal preprocessing is applied to the field images. Both "close-up" and "landscape" images are resized to a fixed size defined by the backbone architecture, and z-score normalized [48] based on the mean and standard deviation of the training data for each input channel, i.e.,

$$X_{\text{norm}}^{(P)} = \frac{X^{(P)} - \mu}{\sigma}. \tag{3}$$

The outputs of this step, aligned, harmonized satellite data and field images, serve as inputs to the multimodal pseudo-labeling framework described next.

B. Multimodal Pseudo-Labeling

The second step of the method aims to automatically generate an extra set of pseudo labels that can mitigate the lack of reference data. To this end, we leverage the complementary information provided by the considered data sources. Fig. 2 shows the proposed implementation, where outputs of unimodal encoders are combined using a fusion objective. The resulting multimodal classifier is then used to predict pseudo crop labels. These labels are referred to as pseudo because, unlike field-collected labels, they are generated by a classifier and may not be reliable.

Given the recent developments in crop-type mapping literature [11], [19], we use a transformer-based architecture as

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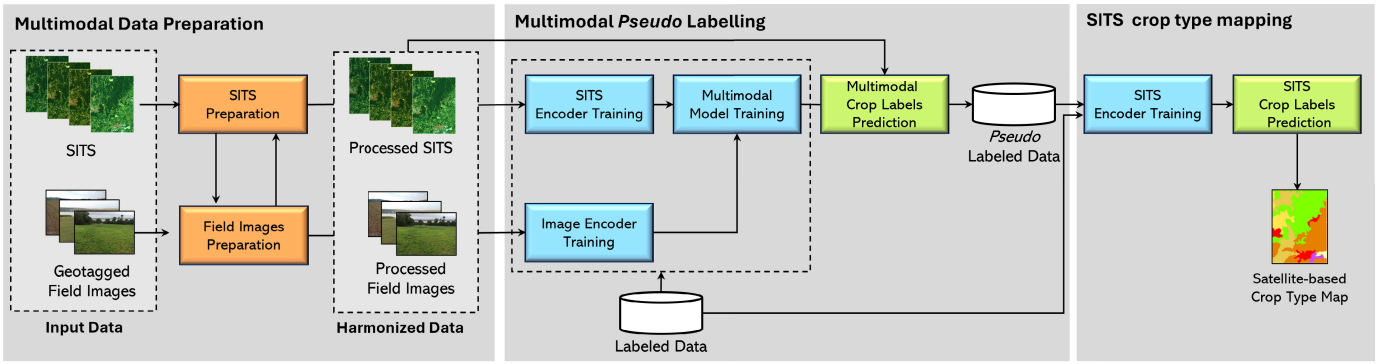


Fig. 1. Proposed multimodal pseudo-labeling approach for enhanced crop-type mapping. The method is made up of three main steps: 1) multimodal data preparation; 2) multimodal pseudo-labeling; and 3) SITS-based crop time mapping.

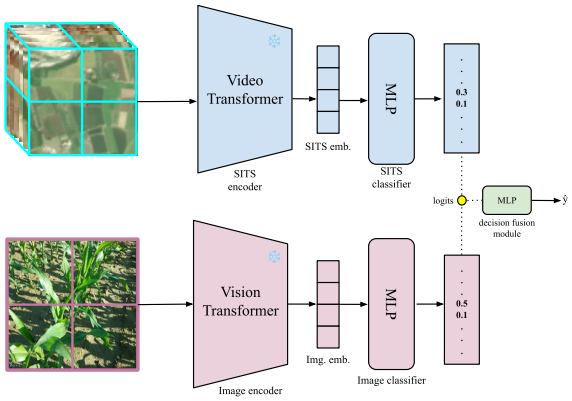


Fig. 2. Multimodal model training. Each modality is processed by a separate encoder before the considered fusion step.

our SITS encoder [49]. In particular, we chose a video vision transformer (VViT) [50], which allows us to process both spatial and temporal dimensions of the input. Although initially developed for video analysis in computer vision applications, the monthly composites of SITS result in uniform temporal intervals, which makes VViT suitable for their classification. This approach is widely adopted in the literature for crop-type mapping using SITS [47], [49], [51]. The VViT consists of two standard transformer encoders, a spatial transformer block, and a temporal transformer block applied in sequence. A standard vision transformer (ViT) [52] is considered for the image encoder. As described by the original authors, this approach involves splitting an image into fixed-size patches, applying a linear embedding to each patch, adding positional encodings, and feeding the resulting sequence of vectors into a standard transformer encoder. This allows us to benefit from existing models pretrained on large datasets of natural images. In the considered setup, we compare a ViT large (ViT-L), which offers a good tradeoff regarding parameters and accuracy [53], and a ViT base (ViT-B) as a more computationally efficient architecture. We provide specific details on each architecture in the Appendix.

We first train a multimodal classifier g using the initial labeled multimodal dataset $\mathcal{D}$, evaluating and comparing decision fusion, late fusion, and ensemble approaches [18], [54]. In the decision fusion strategy, separate classifiers are trained for each modality, generating individual predictions $\hat{y}^{(m)} \in \mathbb{R}^K$. The final prediction $\hat{y}$ is obtained by aggregating the outputs of the unimodal classifiers using a fusion function, $\text{merge}(\{\hat{y}^{(m)}\}_{m\in\mathcal{M}})$. Ensemble aggregation operates in two steps, unlike decision fusion. First, each classifier is trained independently. Then, the predicted probabilities from each classifier are aggregated to obtain the final prediction. Therefore, this method does not require a multimodal training step. In late fusion, each modality $m \in \mathcal{M}$ is processed by a modality-specific encoder $f^{(m)}$ that extracts feature representations $z^{(m)}$ that are then merged to produce a joint representation $z = \text{merge}(\{z^{(m)}\}_{m\in\mathcal{M}})$. Then, a learnable classification layer processes the joint representation and outputs the final prediction. Compared with alternatives to late multimodal approaches, it typically allows for the best flexibility as it explicitly optimizes over multimodal representations.

Finally, we use the trained classifier to generate pseudo-labels. Pseudo-labels are denoted by $\hat{y}$, for the unlabelled multimodal dataset $\mathcal{D}' = \{(X_i^{(S)}, X_i^{(P)})\}_{i=1}^{N'}$. These pseudo-labels are then used to create an augmented labeled dataset $\mathcal{D}_{\text{concat}} = \mathcal{D} \cup \{(X_i^{(S)}, X_i^{(P)}, \hat{y}_i)\}_{i=1}^{N'}$. This enriched training set also increases the spatial coverage for the training data.

C. SITS Crop-Type Mapping

In the last step of the method, we generate the crop-type map of the entire study area leveraging the information provided by the SITS and the enriched training set $\mathcal{D}_{\text{concat}}$. Although multimodality may allow us to increase the crop classification accuracy, the field images are not available everywhere. Therefore, this step retrains the SITS classifier $f'^{(S)}$ on the augmented dataset $\mathcal{D}_{\text{concat}}$ and uses it to generate the final crop-type map. In the considered implementation, we employed the VViT architecture to achieve results consistent with those obtained in the previous step. However, due to the higher number of training samples, we expect a significant increase in the reliability of the resulting crop-type map.

IV. Dataset Description and Experimental Setup

A. Dataset Description

To evaluate the method's effectiveness, we used the publicly available Sen4Map dataset [49]. This dataset includes SITS composed of 64 × 64 patches of multispectral Sentinel-2 images acquired in 2018, covering over 300 000 geotagged

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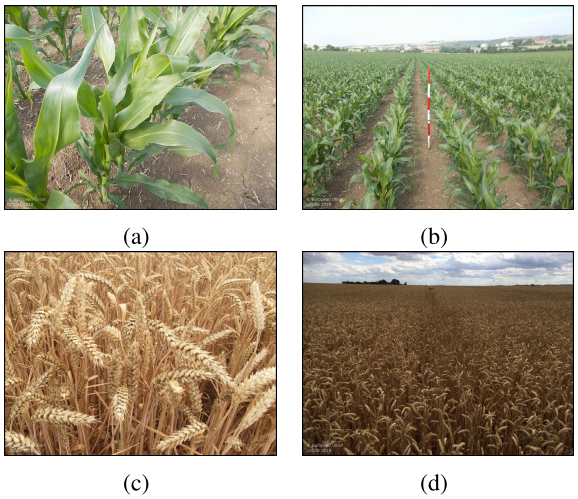


Fig. 3. Examples of LUCAS images. (a) and (b) Maize, (c) and (d) wheat, with (a) and (c) as "close-up" and (b) and (d) as "landscape."

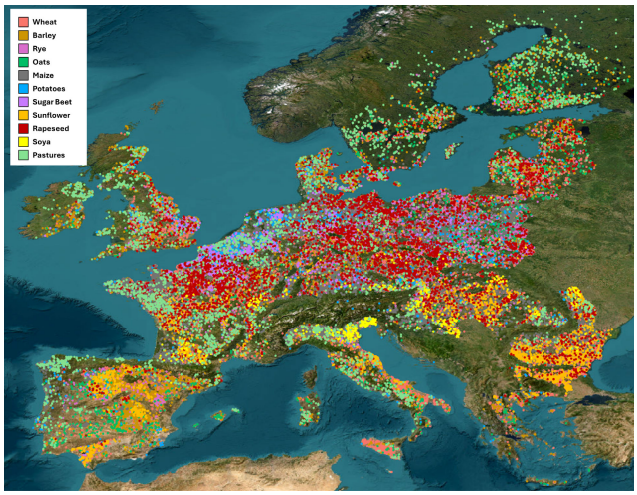


Fig. 4. Spatial distribution of the 2018 LUCAS samples associated with the main crop types present in the EU.

locations across the European Union (EU). The field data collection was conducted as part of the LUCAS initiative, a triennial ground survey organized by the Statistical Office of the European Commission (Eurostat) since 2001, aimed at collecting land-use and land-cover data [55]. Along with standard manual annotations, the survey includes acquiring five "landscape" images captured at the survey point and in the four cardinal directions, and a "close-up" image [56]. Fig. 3 shows two examples of "close-up" [Fig. 3(a) and (c)] and "landscape" images [Fig. 3(b) and (d)] acquired in the field for two different crop types.

For each parcel, we created a custom dataset by selecting patches of size 15 × 15, as outlined in [49] for the crop-type mapping task. Of the 300 000 samples available in the 2018 LUCAS survey, we focused on 45 000 associated with the main cultivation present in the EU, namely, "wheat," "barley," "rye," "oats," "maize," "potatoes," "sugar beet," "sunflower," "rape/turnip," "soya," and "pastures." Fig. 4 represents the spatial distribution of the dataset. In the considered operational scenario, no crop boundary masks were used to isolate pixels specific to the crop. Instead, the entire patch was input into the VViT, significantly reducing the dependence on ancillary information (e.g., crop boundary data) and minimizing data preparation efforts. To carry out the experimental analysis, we focus only on the "close-up" image and the "landscape" image taken at the actual point (P). Ancillary images captured in the four cardinal directions, i.e., east, north, south, and west, were not considered.

B. Experimental Setup

Our experimental setup involves multimodal datasets with varying sample sizes to assess data quality and complexity. The considered multimodality strictly refers to the top and the bottom view perspectives provided by the satellite data and the field images. To this end, we took advantage of the availability of two subsets released as part of the harmonized LUCAS dataset [45], [57], i.e., the LUCAS Cover and LUCAS Copernicus. The LUCAS Cover is a subset of LUCAS field images manually selected to ensure a clear view of mature, preharvest crops, free from obstructions or foreign objects [56]. This results in a dataset of 3532 images acquired in 2018, split into training/validation (1245 parcels) and testing (2287 parcels). This dataset allows robust model training and a reliable evaluation of field image classification performance for our benchmark analysis. The curated training set is hereafter referred to as LUCAS Point-1k (P1K) and LUCAS Cover-1k (C1K) for the "landscape" and "close-up" images, respectively. In contrast, the LUCAS Copernicus subset includes approximately 16 000 crop samples, where each LUCAS theoretical point location falls within manually delineated LUCAS polygons. This dataset is randomly split into training (13.600 parcels) and validation (2.400 parcels) subsets. These data are referred to as LUCAS Point-16k (P16k) and LUCAS Cover-16k (C16k) for the "landscape" and "close-up" images, respectively. To further validate the VViT model trained using the Sentinel-2 SITS, we construct a larger training set without modality pairing constraints by considering all the 45 000 crop samples available in the 2018 LUCAS dataset.

V. Experimental Results

A. Field Images

The first set of experiments aims to assess the potential of the field images by comparing the results obtained when considering: 1) image modality, i.e., "landscape" and "close-up" images; 2) DL models of different sizes; and 3) the number of labeled samples required. We report results on "close-up" images and "landscape" images in Table I. For reference, it is worth mentioning that the convolutional neural network (CNN) baseline provided by [56] on the curated LUCAS Cover dataset made up of 15 876 field images from 2006 to 2018 achieves 75% Macro-$F1$ on their full unbalanced test set.

To extensively test the multimodal interactions, the implemented framework adopts a plug-and-play approach, which allows the selection of encoders that best suit specific needs. As such, we rely for comparison on widely available off-the-shelf encoders. Specifically, we compared the CNNs [58] with the ViTs [52] for the field image encoder and

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TABLE I

CLASSIFICATION RESULTS OF UNIMODAL ENCODERS ON THE CURATED TESTING SET MADE OF 2287 "CLOSE-UP" AND "LANDSCAPE" IMAGES, RESPECTIVELY, OBTAINED BY TRAINING THE MODEL WITH BOTH THE "LANDSCAPE" AND THE "CLOSE-UP" IMAGES. C1K: MODEL TRAINED ON 1245 "CLOSE-UP" IMAGES BELONGING TO THE CURATED DATASET, C16K: MODEL TRAINED ON 13.600 "CLOSE-UP" IMAGES BELONGING TO THE COPERNICUS DATASET, AND P16K: MODEL TRAINED ON 13.600 "LANDSCAPE" IMAGES BELONGING TO THE COPERNICUS DATASET

Class	Sup.	Prediction on 2287 "close-up" images									Prediction on 2287 "landscape" images								
		ResNet50			ViT-B			ViT-L			ResNet50			ViT-B			ViT-L		
		C1K	C16K	P16K	C1K	C16K	P16K	C1K	C16K	P16K	C1K	C16K	P16K	C1K	C16K	P16K	C1K	C16K	P16K
Wheat	270	77.9	70.6	21.5	70.7	73.4	20.9	**87.1**	83.8	73.1	46.5	36.9	54.7	35.0	42.5	53.0	44.2	45.8	**64.0**
Barley	235	60.0	61.4	25.4	58.4	66.5	16.4	75.5	**78.6**	44.3	20.5	32.0	48.9	4.1	36.7	42.1	33.8	38.3	**63.5**
Rye	120	49.2	42.2	31.9	47.4	49.8	32.4	69.7	**72.7**	42.5	28.1	19.0	30.0	25.2	23.0	28.1	34.1	32.9	**42.6**
Oats	152	76.5	72.1	17.3	69.3	65.9	2.4	92.8	**93.6**	35.4	19.4	22.8	36.6	17.7	20.8	31.2	42.2	39.4	**50.9**
Maize	331	94.8	89.8	50.0	95.5	93.6	52.1	**99.7**	99.4	95.4	64.6	59.8	83.6	87.6	67.3	90.6	95.9	94.6	**96.3**
Potatoes	212	83.2	80.2	16.0	81.7	84.0	25.9	**98.6**	97.4	93.4	53.1	40.1	57.1	60.2	59.6	67.8	88.3	81.8	**89.1**
Sugar beet	196	89.9	93.9	21.8	91.8	92.0	42.8	**99.2**	99.0	98.7	31.4	48.4	67.3	40.8	37.1	76.1	89.5	88.0	**92.5**
Sunflower	44	86.0	62.4	16.6	91.3	67.2	18.0	**100.0**	96.7	53.0	23.9	33.5	48.4	74.5	34.4	54.1	**87.2**	71.4	77.8
Rape	226	94.5	88.2	59.2	91.3	88.1	64.8	**99.1**	97.8	94.3	63.3	63.0	70.7	53.0	51.8	76.4	77.4	76.9	**86.3**
Soya	135	82.8	67.6	0.0	81.2	69.0	4.1	**98.5**	93.4	62.0	10.2	7.0	9.5	8.9	4.2	47.1	74.8	51.0	**75.6**
Pastures	366	86.2	86.4	52.1	86.7	87.2	47.7	**97.1**	95.4	84.6	57.8	57.3	69.3	57.1	61.0	71.0	74.3	68.3	**87.2**
Macro-F1 (%)		80.1	74.1	28.3	78.7	76.1	29.8	**92.5**	91.6	70.6	38.1	42.2	67.4	38.2	39.9	62.6	52.4	57.9	**75.1**
Accuracy (%)		82.4	78.3	35.9	80.4	80.0	36.7	**93.0**	92.2	76.9	46.5	48.9	69.7	45.0	48.3	65.3	60.5	63.5	**78.5**
Kappa (%)		80.2	75.6	28.4	74.8	74.4	34.1	**86.5**	85.7	71.5	39.9	45.5	64.8	37.8	44.9	60.7	55.4	59.1	**73.0**
Train Param. (M)			0.5			0.2			0.3			0.5			0.2			0.3	
Total Param. (M)			24.0			86.0			304.4			24.0			86.0			304.4	

finetune existing pretrained models with multilayer perceptron (MLP) probing.

In our experiments, ViTs and CNNs with comparable parameters achieve similar classification performance. As expected, larger models perform better, with the ViT-L model consistently outperforming the ViT-B model, regardless of the training sample size. However, it is important to note that the image modality also influences the results obtained. Models trained on "close-up" images perform better than those trained on "landscape" images when evaluated on the test "close-up" images and vice-versa. On the other hand, cross-domain shifts affect "close-up" trained models more. As expected, transferring "close-up" models to landscapes results in a larger performance drop than transferring from "landscape" to "close-up." Indeed, models trained on "landscape" images learn to focus on more large-scale crop structures, which can still be partially retrieved in "close-up" images. In contrast, "close-up" trained models rely on more discriminative fine-grained features that may not be visible at the landscape scale.

B. Fusion Comparison

In this section, we present the comparison of the results obtained with the considered fusion strategies, i.e., the late fusion, decision fusion, and ensemble aggregation strategies, typically used in the literature [59]. For each late-fusion method, we rely on a simple merging function. For ensemble aggregation, output logits are mean aggregated. For decision-fusion, output logits are aggregated based on the learned weighting scheme. Finally, for late-fusion, high-level modality-specific representations are concatenated before the final classification layer.

According to the results presented in Section V-A, the best modality for collecting field images is the "close-up." Therefore, in this set of experiments, we integrate our SITS encoder (VViT) with a "close-up" image encoder ViT-B or ViT-L to evaluate the impact of the fusion strategy.

Table II presents the fusion results obtained from training the models on the 16k and 1k datasets, respectively.

From these results, one can notice that consistent with the results presented in Section V-A, better unimodal encoder performance leads to better multimodal performance overall. As expected, a larger ViT-L encoder outperforms ViT-B. Focusing on the ViT-L model, the results indicate that decision fusion is the most effective strategy to combine data sources, followed by ensemble aggregation and late fusion. In particular, decision fusion achieves consistently high performance with both the C1K and C16k datasets despite the difference in the number of training samples, yielding a Macro-$F1$ of 90.1% and 91.6%, respectively. In addition, accurate classification results are achieved across the different crop types regardless of the number of samples available per class, having a minimum $F1$ of 60.5% for the "rye" class and 99.0% for "sugar beet" and "maize."

Focusing on the ViT-B model, the decision fusion strategy yields the best results on the 16k dataset, achieving a Macro-$F1$ score of 80.6%, followed by ensemble fusion (79.5%) and late fusion (72.9%). In contrast, ensemble fusion performs best on the 1k dataset with a Macro-$F1$ of 77.9%, followed by decision fusion (70.2%) and late fusion (66.7%). The lowest $F1$ score for both datasets is observed for the "rye" class, with 25.5% in decision fusion and 30.4% in late fusion.

C. SITS Crop-Type Mapping

To generate the pseudo-labels, we apply the same preprocessing steps to the new input data as during training and perform inference using the considered multimodal classifier. Based on the results presented in Section V-B, we focus

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TABLE II

CLASSIFICATION RESULTS ON THE CURATED TEST SET (2287 MULTIMODAL SAMPLES) AND TRAINED ON THE 16K AND THE 1K DATASET, RESPECTIVELY. THE VVIT MODEL TRAINED ON THE SITS IS COMBINED WITH EITHER VIT-B OR VIT-L, BOTH TRAINED ON "CLOSE-UP" IMAGES. THREE FUSION STRATEGIES ARE COMPARED: LATE FUSION, DECISION FUSION, AND ENSEMBLE AGGREGATION

Class	Support	Model trained on the 16K dataset						Model trained on the 1K dataset					
		VViT + ViT-B (C16K)			VViT + ViT-L (C16K)			VViT + ViT-B (C1K)			VViT + ViT-L (C1K)		
		Late	Decision	Ensemble	Late	Decision	Ensemble	Late	Decision	Ensemble	Late	Decision	Ensemble
Wheat	270	65.1	74.2	71.5	79.0	**84.4**	80.4	61.1	64.5	69.3	73.8	**84.6**	78.5
Barley	235	59.0	70.0	68.0	72.3	**81.6**	77.2	33.0	40.4	59.6	54.5	**73.2**	62.9
Rye	120	32.1	44.0	36.1	51.4	**70.4**	63.8	30.4	25.5	45.5	40.6	**60.5**	55.4
Oats	152	41.1	67.5	55.7	80.4	**91.5**	86.9	41.5	52.8	61.0	71.2	**89.9**	78.2
Maize	331	94.8	96.7	96.7	98.9	99.1	**99.2**	91.2	92.6	95.7	98.2	**99.7**	94.5
Potatoes	212	82.6	88.2	91.8	97.4	97.0	**98.6**	72.6	70.7	82.8	97.4	**98.3**	93.0
Sugar beet beet	196	95.3	96.1	96.4	**99.0**	99.0	99.0	89.4	87.6	93.2	99.5	**99.7**	96.4
Sunflower	44	76.4	81.9	85.1	97.8	97.8	**98.9**	69.1	84.8	86.0	**98.9**	96.6	87.9
Rape	226	92.6	95.6	96.1	**98.7**	98.5	98.3	83.7	91.3	94.0	**99.1**	98.2	95.5
Soya	135	71.7	78.9	85.1	**94.6**	91.3	**94.6**	76.4	76.2	80.1	94.2	**95.6**	88.8
Pastures	366	91.0	92.9	91.7	96.0	**96.7**	95.0	85.0	85.4	89.5	94.5	**94.8**	91.3
Macro-F1 (%)		72.9	80.6	79.5	87.8	**91.6**	90.2	66.7	70.2	77.9	83.8	**90.1**	83.9
Accuracy (%)		76.5	84.6	83.5	92.2	**96.1**	94.7	70.0	73.7	81.8	88.0	**94.6**	88.1
Kappa (%)		71.2	78.7	77.6	85.7	**89.4**	88.0	65.1	68.5	76.1	81.8	**88.0**	81.9

TABLE III

CLASSIFICATION RESULTS OBTAINED ON THE SET OF SAMPLES USED TO GENERATE THE PSEUDO-LABELS AND TESTING SET USING VIT-L AND VVIT TRAINED ON THE 16K DATASET AS WELL AS THE PROPOSED METHOD (PM)

Class	Parcels for *pseudo*-label generation				Test Set (VViT)				
	Support	ViT-L (16K)	VViT (16K)	Fusion	Support	1K	16K	45K	PM
Wheat	8723	84.1	74.6	**86.7**	271	53.5	58.5	**65.0**	58.9
Barley	4094	71.4	53.0	**74.6**	235	20.2	46.2	**52.0**	48.0
Rye	981	58.7	39.3	**61.8**	121	20.6	20.4	**26.1**	24.3
Oats	1102	62.4	22.5	**64.6**	152	**27.3**	16.1	23.1	16.9
Maize	5650	96.4	88.7	**97.9**	333	73.6	80.2	83.3	**85.4**
Potatoes	665	89.3	55.0	**91.1**	214	47.7	61.8	**76.5**	75.5
Sugar beet	651	92.2	78.8	**95.1**	197	70.5	86.3	89.9	**91.3**
Sunflower	1192	92.8	78.9	**95.2**	44	43.4	61.5	**76.9**	60.5
Rape	2671	88.9	80.9	**93.8**	228	79.8	89.8	91.3	**92.1**
Soya	231	83.8	49.0	**89.4**	135	56.3	62.7	75.4	**78.8**
Pastures	1737	82.1	71.0	**85.7**	366	73.2	79.4	**81.3**	78.9
Macro-F1 (%)		82.0	62.9	**85.1**		51.5	60.3	**67.4**	64.6
Accuracy (%)	27697	84.1	71.5	**86.9**	2287	57.6	66.7	**72.2**	70.6
Kappa (%)		80.5	64.8	**83.9**		49.4	62.4	**68.7**	66.8

on the best-performing methods, combining the ViT-L and VViT models trained with 16k samples using a decision fusion strategy. The results, shown in Table III, include the accuracy achieved on the set of samples used to generate the pseudo-labels (27 697 parcels from the original 45k, after excluding the 16k used for training and considering only those with both modalities available). In addition, Table III presents the accuracy on the curated test set of 2287 samples when training the VViT using only SITS data with 1k, 16k, and 45k labeled training samples, as well as with the proposed enriched training set (PM), which consists of 16k labeled samples and 27 697 pseudo-labels generated through the multimodal approach. In particular, the VViT model trained using all 45k labeled samples is used as upper bound performance to evaluate the effectiveness of the proposed pseudo-label dataset.

As expected, the ViT-L model achieved lower accuracy on the field images that were not manually inspected than on the curated dataset (e.g., the test set). Indeed, these 27 697 parcels may include images collected during incorrect seasons (e.g., immediately after harvest) or reporting foreign objects that can affect their classification accuracy. However, these results are still relevant to improve the performance obtained when using the VViT model alone, thus leading to a minimum $F1$ score of 58.7% for the "rye" class and a maximum of 96.4% for "maize," compared with a minimum $F1$ of 22.5% for "oats" and a maximum of 88.7% for "maize" when relying solely on SITS data trained with 16k samples. Furthermore, the multimodal approach improves classification accuracy across all crop types, highlighting the value of integrating satellite top-view data with ground-level field images.

Focusing on the testing set results, performance improves as the samples increase from 1k to 45k, as expected. However, given the complexity of the crop-type mapping task, some classes, like "oats" ($F1$: 23.1%), show lower $F1$ scores even when using the whole amount of data available. This is primarily due to the European-scale analysis with a relatively small sample size compared to existing literature [47], [51], and the detailed classification scheme [49]. Moreover, these

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results align with those achieved in the literature when using the 2018 LUCAS data to produce a continental-scale crop-type map with a similar number of samples [60]. Notably, the proposed approach achieves performance comparable to using the entire labeled dataset (45k) while significantly reducing the need for field-collected labeled data. In particular, we observe that retraining the SITS model with additional pseudo-labeled samples improves its overall Macro-$F1$ score to 64.6%, compared with 60.3% for the original model trained with the same amount of labeled data (16k). This performance is also close to the 67.4% achieved using the available labeled dataset. Notably, crops with fewer samples in our dataset benefit from this approach: for instance, the $F1$ score for "soya" increased from 62.7% to 78.8%, while "potatoes" improved from 61.8% to 75.5%.

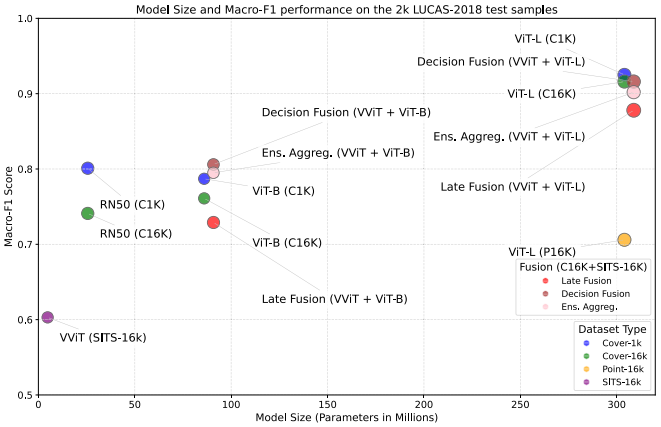


Fig. 5. Comparative $F1$ scores for crop classification obtained on the curated LUCAS test samples (2287 samples) using unimodal models and the different fusion strategies.

VI. Discussion and Conclusion

This article presents an approach to large-scale crop classification that integrates multimodal information from SITS and field images to address the challenge of labeled data scarcity. By integrating the considered multimodal data, the proposed method reduces the need for costly labeled data while maintaining performance comparable to models trained on fully labeled datasets. In addition, we compared different field image collection modalities, i.e., "landscape" and "close-up", to determine which provides the most valuable information for crop-type classification. This article also benchmarks DL models of various sizes and configurations to classify field images. Finally, we evaluated three standard data fusion approaches, i.e., late, decision, and ensemble.

Fig. 5 shows the comparison of the performance achieved with the three fusion strategies versus their unimodal alternatives when considering different model sizes and different amounts of training data. The results highlight the potential of multimodal data fusion in boosting model accuracy. From the results obtained, one can observe the composition in the field image modality matters. In our experiments, models trained on "close-up" images outperformed those trained on "landscape" images, and vice-versa, when evaluated in-domain. Although less discriminative, "landscape" trained models exhibited better robustness to cross-domain shifts, and the contextual information they provide should not be discarded; for example, they offer better localizability [61], [62]. In addition, we see that better field-level image quality, e.g., using curated data, improves training sample efficiency.

Our experiments show that simple decision-level fusion outperforms unimodal alternatives with minimal increase in the model parameters. We argue that the field-level modality brings a complementary viewpoint, which allows for better disambiguation. Therefore, we propose a multimodal pseudo-labeling pipeline to increase the quality and the number of annotations for supervised SITS models. The results demonstrate that the multimodality allows for improved crop-type classification. Indeed, models trained on the pseudo-labeled dataset achieved performance comparable to models trained with ground-truth labels.

The limitation of the proposed framework is that it requires: 1) a set of high-quality, geo-referenced, and time-stamped field images for multimodal training and 2) an initial training dataset that adequately represents the class distribution. Indeed, if the initial training data is biased or insufficient, any pseudo-labels derived from it may propagate and amplify these biases, potentially degrading model performance. However, this is a common challenge in approaches designed to expand labeled data and enhance the training of supervised classification models, such as semi-supervised methods. Both assumptions are reasonable from an operational perspective. Field images are now collected alongside reference data during field campaigns, making it feasible to assume access to high-quality field images paired with in situ data that accurately represents the class distribution. Moreover, in regions with limited field data, the proposed method can incorporate lower quality crowdsourced images. The results indicate that when trained on high-quality field images, the model exhibits robustness to noisy data when applied to a large dataset of field images. Finally, at extreme phenological stages (e.g., seeding or postharvest), images may provide limited visual cues for direct crop-type classification, which needs to be taken into account. As future developments, we aim to investigate other fusion strategies and learning paradigms. In particular, multimodal self-supervised learning is naturally suited for aligning field images with abundant unlabelled SITS. In addition, we aim to expand our approach by incorporating ancillary data sources, such as ecoregion information, climate and soil data or crop calendars, to enrich the multimodal framework further. Finally, we aim to explore the potential of multimodality to improve the temporal and spatial generalization capability of the crop-type mapping models, which are often challenged by interannual variability in crop phenology and cultivation patterns driven by climate fluctuations and crop rotation. To this end, we aim to test the proposed approach on the 2022 LUCAS database, paired with 2022 Sentinel-2 SITS, and investigate the possibility of expanding the classification scheme by including more crop types.

Appendix A
Additional Results

Table IV shows a comparison of existing baseline methods for the classification of the Sentinel-2 SITS when

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TABLE IV
UNIMODAL SITS CLASSIFICATION RESULTS ON THE LUCAS 2018 TESTING SET

1k Training Samples		Transformer			Inception			LSTM			TCN			VViT		
Class	S	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1
Wheat	270	43.1	67.4	52.6	46.7	63.0	53.6	45.1	67.8	54.1	47.9	54.1	50.8	48.1	60.4	53.5
Barley	235	30.6	25.1	27.6	33.6	17.9	23.3	44.3	20.0	27.6	32.5	33.2	32.8	31.5	14.9	20.2
Rye	120	33.3	4.2	7.4	21.7	23.3	22.5	8.5	5.0	6.3	18.2	8.3	11.4	19.0	22.5	20.6
Oats	152	30.4	25.0	27.4	27.3	32.9	29.9	23.9	36.8	29.0	26.5	40.1	31.9	36.7	21.7	27.3
Maize	331	77.6	50.2	60.9	79.6	67.4	73.0	72.5	65.3	68.7	71.1	65.6	68.2	79.9	68.3	73.6
Potatoes	212	38.8	44.3	41.4	54.1	52.8	53.5	57.0	50.0	53.3	61.6	25.0	35.6	61.0	39.2	47.7
Sugar beet	196	84.7	70.4	76.9	88.0	63.8	74.0	77.7	74.5	76.0	69.2	80.1	74.2	67.0	74.5	70.5
Sunflower	44	25.9	68.2	37.5	30.6	59.1	40.3	44.6	56.8	50.0	25.7	61.4	36.2	37.1	52.3	43.4
Rape	226	91.4	70.8	79.8	80.7	77.9	79.3	77.8	68.1	72.6	85.5	73.0	78.8	74.0	86.7	79.8
Soya	135	33.2	74.1	45.9	59.4	70.4	64.4	54.0	54.8	54.4	57.2	61.5	59.3	43.0	81.5	56.3
Pastures	366	71.3	62.6	66.7	66.2	71.3	68.7	67.6	75.1	71.2	66.2	71.6	68.8	71.4	75.1	73.2
Macro F1 (%)		47.6			52.9			51.2			49.8			51.5		
Accuracy (%)		52.5			57.2			56.3			55.1			57.6		
Kappa (%)		47.1			52.1			51.0			49.7			49.4		
Parameters (M)		0.2			0.1			1.3			1.0			1.2		
16k Training Samples		**Transformer**			**Inception**			**LSTM**			**TCN**			**VViT**		
Class	**S**	**P**	**R**	**F1**	**P**	**R**	**F1**	**P**	**R**	**F1**	**P**	**R**	**F1**	**P**	**R**	**F1**
Wheat	270	37.9	82.2	51.9	41.0	79.6	54.1	37.9	79.3	51.3	39.0	81.1	52.6	44.8	84.4	58.5
Barley	235	40.6	45.1	42.7	39.7	40.0	39.8	37.1	43.8	40.2	40.0	40.0	40.0	43.7	48.9	46.2
Rye	120	30.4	14.2	19.3	44.0	9.2	15.2	35.1	10.8	16.6	38.6	14.2	20.7	55.6	12.5	20.4
Oats	152	31.2	3.3	6.0	27.0	6.6	10.6	31.6	3.9	7.0	36.4	10.5	16.3	34.0	10.5	16.1
Maize	331	72.9	77.9	75.3	58.2	88.8	70.3	64.8	80.7	71.9	62.8	86.1	72.6	70.8	92.4	80.2
Potatoes	212	83.2	42.0	55.8	80.2	47.6	59.8	76.8	34.4	47.6	72.7	41.5	52.9	76.4	51.9	61.8
Sugar beet	196	87.2	80.1	83.5	86.0	78.1	81.8	77.5	75.5	76.5	86.9	78.1	82.3	89.1	83.7	86.3
Sunflower	44	43.8	72.7	54.7	43.9	65.9	52.7	34.8	70.5	46.6	44.8	68.2	54.1	53.3	72.7	61.5
Rape	226	81.1	87.2	84.0	77.9	82.7	80.3	83.8	73.5	78.3	82.5	79.2	80.8	90.2	89.4	89.8
Soya	135	85.6	61.5	71.6	93.2	50.4	65.4	83.0	57.8	68.1	88.2	49.6	63.5	81.2	51.1	62.7
Pastures	366	75.9	65.3	70.2	83.3	62.6	71.5	80.0	67.8	73.4	79.7	63.4	70.6	86.2	73.5	79.4
Macro F1 (%)		55.9			54.7			52.5			55.1			60.3		
Accuracy (%)		61.4			60.8			58.9			60.3			66.7		
Kappa (%)		56.5			55.7			53.6			55.2			62.4		
Parameters (M)		0.2			0.1			1.3			1.0			1.2		
45k Training Samples		**Transformer**			**Inception**			**LSTM**			**TCN**			**VViT**		
Class	**S**	**P**	**R**	**F1**	**P**	**R**	**F1**	**P**	**R**	**F1**	**P**	**R**	**F1**	**P**	**R**	**F1**
Wheat	270	39.0	83.3	53.1	36.7	84.8	51.2	35.2	84.8	49.8	39.3	84.8	53.7	52.3	85.9	65.0
Barley	235	47.3	36.6	41.2	48.7	40.0	43.9	45.3	34.5	39.1	44.3	43.4	43.9	49.8	54.5	52.0
Rye	120	53.3	20.0	29.1	52.9	7.5	13.1	44.4	6.7	11.6	56.5	10.8	18.2	51.2	17.5	26.1
Oats	152	40.3	19.1	25.9	44.7	13.8	21.1	33.3	12.5	18.2	42.6	15.1	22.3	39.1	16.4	23.1
Maize	331	64.8	88.8	74.9	66.2	88.2	75.6	65.6	88.2	75.3	69.0	87.6	77.2	75.2	93.4	83.3
Potatoes	212	82.9	54.7	65.9	84.6	51.9	64.3	76.0	46.2	57.5	85.1	48.6	61.9	79.6	73.6	76.5
Sugar beet	196	88.3	80.6	84.3	88.1	79.1	83.3	83.9	74.5	78.9	88.4	78.1	82.9	91.1	88.8	89.9
Sunflower	44	50.0	68.2	57.7	51.6	75.0	61.1	35.6	70.5	47.3	41.7	79.5	54.7	74.5	79.5	76.9
Rape	226	90.6	85.0	87.7	87.7	85.0	86.3	88.7	79.6	83.9	90.1	81.0	85.3	87.4	95.6	91.3
Soya	135	92.5	54.8	68.8	97.3	53.3	68.9	88.6	51.9	65.4	92.5	54.8	68.8	92.5	63.7	75.4
Pastures	366	82.5	64.5	72.4	81.5	67.2	73.7	84.6	61.5	71.2	80.1	69.1	74.2	90.9	73.5	81.3
Macro F1 (%)		60.1			58.4			54.4			58.5			67.4		
Accuracy (%)		64.0			63.5			60.3			63.8			72.2		
Kappa (%)		59.4			58.8			55.2			59.1			68.7		
Parameters (M)		0.2			0.1			1.3			1.0			12.6		

Note: Values presented as Precision / Recall / F1 (%). S = Support. Best results are highlighted.

using the 1k, 16k, and 45k training datasets. In the table, we report standard classification metrics[1]: overall accuracy is $\mathrm{OA} = (1/n)\sum_{i=1}^{k} TP_i$, precision for class i is $\mathrm{Precision}_i = (\mathrm{TP}_i/\mathrm{TP}_i + \mathrm{FP}_i)$, recall for class i is $\mathrm{Recall}_i = (\mathrm{TP}_i/\mathrm{TP}_i + \mathrm{FN}_i)$, F_1-score for class i is $F_{1,i} = (2\,\mathrm{TP}_i/2\,\mathrm{TP}_i + \mathrm{FP}_i + \mathrm{FN}_i)$, Macro-$F_1$ is $(1/k)\sum_{i=1}^{k} F_{1,i}$, and Cohen's kappa is $\kappa = (\mathrm{OA} - P_e/1 - P_e)$, $P_e = (1/n^2)\sum_{i=1}^{k}(\mathrm{TP}_i + \mathrm{FN}_i)(\mathrm{TP}_i + \mathrm{FP}_i)$. Here TP_i, FP_i, FN_i denote true positives, false positives and false negatives; $n = \sum_{i=1}^{k}(\mathrm{TP}_i + \mathrm{FN}_i)$ is the total sample size and k the number of classes. Figs. 6 and 7 report some qualitative examples of classification outputs on the pseudo-labeling set and outputs from fusion models on testing set samples, respectively. Thanks to the complementary information from SITS and field photographs, their integration enables accurate crop classification, even when each data source alone may fail.

[1]https://scikit-learn.org/stable/modules/model_evaluation.html#classification-metrics

APPENDIX B
IMPLEMENTATION DETAILS

A. Unimodal Training Times and Hardware

For 16k and 45k datasets, models were trained using an NVIDIA RTX A4000 GPU. For the Hyperparameters are listed

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Fig. 6. Examples of classification outputs on the pseudo-labeling set. (Class probabilities are in parentheses.)

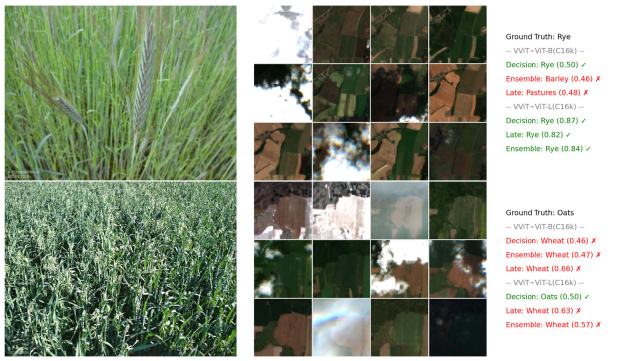


Fig. 7. Examples of classification outputs from fusion models on testing set samples. (Class probabilities are in parentheses.)

TABLE V
SITS MODEL ARCHITECTURES AND HYPERPARAMETERS

Cat.	Parameter	TCN [63]	LSTM [64]	Transformer [65]	InceptionTime [66]	ViT
Data	Input Channels	10	10	10	10	10
	Sequence Length	12	12	12	12	12
	Image Size	–	–	–	–	15×15
Architecture	Embedding Dim	128	128	64	32	1k/16k: 128, 45k: 384
	Hidden/MLP Dim	–	–	128	–	256
	Layers/Depth	–	4	5	2	–
	Spatial Depth	–	–	–	–	1k/16k: 3, 45k: 3
	Temporal Depth	–	–	–	–	1k/16k: 3, 45k: 6
	Dropout	0.18	0.57	0.02	0.2	1k/16k: 0.1, 45k: 0.4
Model-Specific	Kernel Size	7	–	–	[9, 19, 39]	–
	Bidirectional	–	Yes	–	–	–
	Layer Norm	–	Yes	–	–	–
	Attention Heads	–	–	2	–	1k/16k: 4, 45k: 12
	Num. Filters	–	–	–	32	–
	Image Patch Size	–	–	–	–	3
	Frame Patch Size	–	–	–	–	3
Optimization	Optimizer	Adam	Adam	Adam	Adam	AdamW
	Learning Rate	1e-3	1e-3	1e-3	1e-3	2e-4
	Weight Decay	1e-4	1e-4	1e-4	1e-4	–
	LR Scheduler	Cosine Annealing with Warmup				Cosine Annealing with Warm Restarts
	Warmup Epochs	5	5	5	5	–
	Restart Period	–	–	–	–	T_0=20, T_{mult}=2
	Batch Size	64	64	64	64	64
	Max Epochs	100	100	100	100	100
	Early Stopping Patience	10	10	10	10	10

in Tables V and VI. 1k dataset, unimodal baselines were trained using an NVIDIA RTX 4070 GPU. All multimodal models were trained on an NVIDIA A4000 GPU. We used I-/O-bound shared computing resources, so training times are indicative. 1k unimodal baselines can all be trained in under 1 h. For 16k and 45k unimodal SITS, the total training times are comparable between architectures, ranging from 2 to 13 h for the longest run. Fine-tuning ViT-B models took under 2 h, while ViT-L took significantly longer, with the longest run at

TABLE VI
VISION MODEL ARCHITECTURES AND HYPERPARAMETERS

Cat.	Parameter	ResNet50[1]	ViT-B/16[2]	ViT-L/16[2]
Data	Image Size	224×224	224×224	512×512
	Channels	3	3	3
Architecture	Backbone	ResNet50	ViT-B	ViT-L
	Feature Dim	2048	768	1024
	MLP Probe	2-layer, 256 hidden dim, Tanh	2-layer, 256 hidden dim, Tanh	2-layer, 256 hidden dim, Tanh
Model-Specific	Pre-trained Weights	ImageNet	ImageNet	ImageNet SWAG
	Patch Size	–	16×16	16×16
	Frozen Backbone	Yes	Yes	Yes
Optimization	Optimizer	Adam	Adam	Adam
	Learning Rate	1e-4	1e-4	1e-4
	Weight Decay	–	1e-3	1e-3
	Batch Size	32	32	32
	Max Epochs	100	100	100
	Early Stopping Patience	20	20	20

TABLE VII
MULTIMODAL FUSION HYPERPARAMETERS

Cat.	Parameter	Ensemble Aggregation	Decision Fusion	Late Fusion
Architecture	Fusion Level	Prediction	Prediction	Feature
	Input	Model Logits	Model Logits	Feature Vectors
	Merge Operation	Mean	MLP	Concatenation
Fusion Model	Structure	–	2-layer, 256 hidden dim, ReLU	2-layer, 512 hidden dim, ReLU
	Dropout	–	0.2	0.2
	Trainable	No	Yes	Yes
Optimization	Optimizer	–	AdamW	AdamW
	Learning Rate	–	1e-3	1e-4
	Max Epochs	–	100	100
	Early Stop Patience	–	10	10
	Batch Size		32	

Note: All fusion approaches use the pre-trained unimodal models with frozen weights.

around 25 h. For TCN, LSTM, InceptionTime, and (Pixel-) transformer architectures, we reuse the same hyperparameters as in the BreizhCrops [51] benchmark.

B. Multimodal Training

We reuse the pretrained unimodal encoders for multimodal training and inference (Table VII). Ensemble aggregation does not necessitate any additional training. For decision fusion, we train an MLP on the output logits of each unimodal classifier, which took at most 3 h. We also train a classification head on the concatenated unimodal representations for late fusion. The longest run for ViT-L was under 10 h. The overall footprint of the framework can be up to three times the cost of training a single model. However, the flexible choice of encoders and fusion methods allows the selection of the best performance and computational efficiency tradeoffs. All models use pretrained backbones with only the MLP probe layers being fine-tuned[2,3]

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[2]https://pytorch.org/vision/main/models/resnet.html
[3]https://pytorch.org/vision/main/models/vision_transformer.html

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